

The Riemann Hypothesis: Cross-Route Synthesis

Part IV of a Five-Part Series

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Abstract

This paper, Part IV of a five-part series [6, 5, 7, 4], collects three classes of material that do not fit the scope of Parts II (Even Dominance) or III (Spectral Routes): the cross-route synthesis comparing all three active branches A, B, Z; the dormant routes and external lines of the wider landscape; and a self-correcting record of hypothesis corrections during the iterative research process.

The cross-route synthesis identifies a shared Galerkin pattern across branches — each conditional reduction involves a fixed-parameter positive structure that must be promoted to a uniform-limit statement via a growing parameter $N(\lambda)$, $d(\lambda)$, or $m(\lambda)$ — and tabulates the characteristic constants ($C^* = 64\pi^2/3$ as the current Branch A boundary-moment benchmark for $\lambda = 5, 7$, $|\kappa_\infty| \lesssim 10^{-4}$ for Branch B, $g_0(m) \approx 2\pi/\log \gamma_m$ for Branch Z).

Series status (v3.0, 2026-05-26; criticality-lens update 2026-05-31). The series is restructured from a trilogy to a five-part series; the conditional reduction status from v2.2 is unchanged. A 2026 working phase adds Section 3: a *heuristic* criticality lens ($\text{RH} \Leftrightarrow \Lambda = 0$) that strategically re-weights the routes, a sharper localization of the common wall (the analytic continuation across $\Re s = 1$), a computable diagnostic built from the *classical* Laguerre–Pólya/Herglotz criterion, and four honest closures of *concrete constructions* (not routes, and several of them self-corrections of internally proposed constructions). None of these closes RH, refutes a route, or changes the conditional-reduction status.

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1 Introduction

This paper, Part IV of a five-part series, serves as the “miscellaneous and combined” component of the landscape atlas:

Part I [6]: Foundations and Obstructions.

Part II [5]: Even Dominance (Branch A).

Part III [7]: Spectral Routes (Branches B and Z).

Part IV (this paper): Cross-Route Synthesis, dormant routes, external lines, methodical dead ends, hypothesis corrections.

Part V [4]: Conclusio.

The three active branches A (Even Dominance), B (Hadamard Strip / I-1 Normal-Family), and Z (CCM Microcluster) are developed in detail in Parts II and III. This paper identifies the structural patterns they share, surveys the wider landscape of dormant and external routes, and provides a transparent record of hypothesis corrections during the iterative development.

2 Cross-Route Synthesis

2.1 Characteristic Constants

Each active branch produces a characteristic constant or scaling law whose identification anchors the conditional reduction:

Branch	Constant	Scaling law	Theoretical identification
A (Even Dom.)	$C^* = 64\pi^2/3 \approx 210.55$	sLI $\sim C^* \lambda/(NL)$ for the $\lambda = 5, 7$ benchmark; $\lambda = 3$ separate and $\lambda = 11$ open	$16 \int_0^\infty t^2/\sinh^2(t/2) dt$ (reverse-fit candidate; exact collar kernel known)
B (Hadamard)	$ \kappa_\infty(11) \lesssim 10^{-4}$	corrected (transient conflation); sign open	finite-part residual curvature of Foundation-Lock
Z (CCM)	$g_0(m) \approx 2\pi/\log \gamma_m$	log-decreasing	Montgomery–Odlyzko (classical)

Remark 2.1 (Identification status). The constant $C^* = 64\pi^2/3$ for Branch A is an identification candidate for the $\lambda = 5, 7$ benchmark. The 27 May audit shows that the simple endpoint-kernel factorisation is false; the exact collar kernel is $K(t) = e^t \cosh(t)/(2 \sinh^2 t)$, so a rigorous derivation must pass through the eigenmode asymptotics. The $\lambda = 3$ sweep is recorded as a low-bandwidth outlier, while the single $\lambda = 11, N = 200$ point ($C_{\text{eff}} \approx 168$) remains an open discriminator (Part II [5], Remark on derivation candidate). The constant κ_∞ for Branch B is established empirically in the conservative range $[1.6, 1.8] \times 10^{-3}$ (Part III [7]). The gap $g_0(m)$ for Branch Z rests on classical pair-correlation heuristics.

2.2 The Shared Galerkin Pattern

Despite their different analytical mechanisms, all three branches share a common structural pattern:

1. **Fixed-parameter positive structure:** At each finite (λ, N) (or (λ, m) for Branch Z), the relevant quantity is computable and has the correct sign/bound.

2. **Uniform-limit requirement:** The conditional reduction requires promoting the fixed-parameter statement to a uniform assertion as $\lambda \rightarrow \infty$ with a parameter $(N(\lambda), d(\lambda), m(\lambda))$ growing alongside.
3. **Transient vs. converged regime:** The warm-up parameter is branch-dependent. For the GF1/ C^* branch the current diagnostic scale is $\xi = NL/\lambda$, not N/L : the $\lambda = 5, 7$ rows enter the benchmark window near $\xi \approx 50$, whereas the $\lambda = 11, N = 200$ point remains an open pre-convergence discriminator. Other Galerkin branches still track their own resolution ratios.

This pattern is consistent with a *Bernstein–Galerkin conjecture*: the Galerkin truncation of the Weil quadratic form at dimension N converges to the infinite-dimensional form in the relevant operator topology, and the convergence is uniform in λ after a λ -dependent warm-up phase.

Remark 2.2 (Status of the conjecture). The Bernstein–Galerkin pattern is an empirical observation from 5–7 data points per branch. It has not been formalised as a mathematical statement. Its role is heuristic: it provides a mechanistic explanation for why all three branches have similar structural gaps (fixed-parameter $\checkmark$, uniform-limit open).

2.3 Two Isolated Proof Slots

The cross-route analysis identifies two concrete, isolated proof slots that, if resolved, would close one or more branches:

1. **B-1: Uniform log-curvature control from Foundation-Lock.** If $H_\lambda(\delta) \leq K\delta^2$ with $K \leq 1.8 \times 10^{-3}$ is derived from the Foundation-Lock structure, then Theorem B-1 (Part III [7]) gives $A_\lambda \leq 1.0063$, closing Branch B.
2. **GF1: Eigenmode asymptotics + factor-16 identification.** If the exact collar kernel and the $q_a(k)$ eigenmode asymptotics rigorously produce the factor 16 in $C^* = 16 \cdot 4\pi^2/3$, and the $\lambda = 11$ discriminator does not persist as a separate branch, then Theorem GF1 (Part II [5]) provides a boundary-moment scaling benchmark for the non-exceptional Branch A regime; the $\lambda = 3$ branch remains a conservative guardrail.

These two slots are *independent*: closing one does not affect the other. Each is a well-posed analytical problem amenable to standard tools (complex analysis for B-1; Fourier analysis and spectral theory for GF1).

Remark 2.3 (Both slots are margin-type — a strategic caveat). Both slots are *margin/coercivity* statements: a uniform bound that closes a gap. They remain well-posed open problems, and nothing below refutes them. Section 3 adds a *heuristic* criticality lens: since RH (if true) sits at the critical value $\Lambda = 0$ with no buffer (Section 3.1), exact-structural mechanisms (identity, symmetry, spectral realization) deserve *additional* strategic weight alongside these margin-type slots. This is a re-reading of priorities, not a theorem and not a refutation (see the caveat in Theorem 3.1).

3 The Sharpened Common Wall: A Criticality Lens and an Analytic-Continuation Localization

A 2026 working phase re-examined all branches through one question — *what kind of argument can close the gap?* — and localized the shared obstruction more sharply. Everything in this section is a **strategic lens and a diagnostic localization**, not a new closure of RH and not a refutation of any route. Several items are honest self-corrections of internally proposed constructions. The conditional-reduction status of Parts II/III is unchanged.

3.1 A criticality lens (heuristic, not a theorem)

The de Bruijn–Newman constant Λ satisfies $\Lambda \leq 0 \Leftrightarrow \text{RH}$; Rodgers and Tao (*Ann. of Math.* 2020) proved $\Lambda \geq 0$. Hence $\text{RH} \Leftrightarrow \Lambda = 0$: if RH holds, it sits *exactly* at the critical value, with no margin to spare.

Remark 3.1 (Criticality lens — and its essential caveat). This invites a *strategic* dichotomy of proof mechanisms: *margin-seeking* (a uniform gap / coercivity / smallness bound) versus *exact-structural* (an identity, a symmetry, or a spectral realization that forces the conclusion without slack). The heuristic reading is that a mechanism producing a strict uniform margin would be establishing something with a buffer, which sits uneasily with a critical truth at $\Lambda = 0$. **Caveat (essential, to avoid over-claiming)**. The step “a uniform margin would force $\Lambda < 0$ ” is an *analogy*, not a theorem. The branch-specific margins — the even-dominance gap $\lambda_1^+ < \lambda_1^-$, the Hadamard constant κ , the CCM residual smallness — are *not* shown to equal a de Bruijn–Newman statement; Λ and these quantities are different objects. Consequently the lens does *not* refute the well-posed proof slots of Section 2 (Theorem 2.3); it only suggests weighting exact-structural routes *in addition*. It is a re-reading of priorities, not a result.

Under this lens the routes re-sort by *type*, not by viability: the margin-type branches A (A7/AVG gap), B ((ii-a) coercivity), Z (residual smallness) keep their exact kernels as assets, while the exact-structural lines — M (Bost–Connes $\leftrightarrow$ Meyer, with the KMS phase transition at $\beta = 1$ playing the role of built-in criticality) and the global adelic trace route (Connes–Consani scaling site / $\mathbb{F}_1$, where the only missing datum is the Hodge-index positivity of an otherwise-available Lorentzian form) — receive additional strategic weight. The empirical signature is consistent: the even-dominance margin grows like a power law $\sim \lambda^{0.72}$ (a criticality signature, not a buffer). This re-weighting is recorded in the project decision log; it does not alter any route’s formal status.

3.2 The common wall: analytic continuation across $\Re s = 1$

Independently of the lens, the correctly-typed routes share a single concrete endpoint. Three lines of work — the Möbius prime-hub, the Wiener–Hopf factorization, and the Sonin/Toeplitz (Connes–Consani) construction — are each *locally/finitely* controllable but break at *exactly* the same place: the global limit, where the local data must assemble into ζ . The unified description is the Weyl–Titchmarsh function M_∞ of the unbounded limit operator; the missing datum is uniform control of that limit from local data *without* importing ζ .

Remark 3.2 (The wall is a classical fact, here sharply localized). A finite Euler product $\prod_{p \in P} (1 - p^{-s})^{-1}$ has, after the substitution $z = -i(s - \frac{1}{2})$, a pole comb in the upper half-plane at $z_{p,k} = 2\pi k / \log p + i/2$ (on $\Im z = \frac{1}{2}$), with a common pole at $z = i/2$ of order $|P|$; hence its logarithmic derivative is not a Herglotz function. The comb is removed only by ζ 's analytic continuation ($\zeta(0)$ finite; the pole at $s = 1$). This is **Euler-product folklore** (the product converges only for $\Re s > 1$). It is not a new obstruction: it is a *precise localization* of the known wall — the missing datum is the analytic continuation across $\Re s = 1$, i.e. a global/functional-equation phenomenon that finite arithmetic-plus-archimedean data structurally do not supply.

3.3 A computable diagnostic from the classical Laguerre–Pólya/Herglotz criterion

Write $\Xi(z) = \xi(\frac{1}{2} + iz)$ and $M_\Xi = -\Xi'/\Xi$. The chain

$$\text{RH} \iff \Xi \in \text{Laguerre-Pólya} \iff M_\Xi \text{ Herglotz} \iff \frac{(M_\Xi(z) - \overline{M_\Xi(w)})}{z - \bar{w}} \succeq 0$$

is **classical** (Hermite–Biehler; the Laguerre–Pólya characterization of real-rooted entire functions; the positivity $\Re \xi'/\xi > 0$ for $\Re s > \frac{1}{2}$ is Voros' criterion). *We claim no novelty for the criterion itself.* The contribution is a *computable, non-circular test protocol* that turns this classical equivalence into a diagnostic instrument: an Euler-anchor giving M_Ξ from ξ'/ξ in the convergent region $\Re w > 1$ (verified numerically to $\sim 10^{-31}$), a ladder of gates (normalization; tail-tightness; a Stieltjes identity; an off-axis Poisson tomography resolving individual residues), and a detector lemma by which an off-line zero forces a negative 1×1 Pick minor just below the migrated pole. The instrument is empirically validated: it detects off-line zeros and discriminates the true zero spectrum from a smooth density-matched control.

Remark 3.3 (What the diagnostic established on a genuine operator). An independent, non-circular discretization of the Connes–Moscovici prolate operator (arXiv:2112.05500) was run through the full instrument. It reproduces Riemann's zero-counting function $N(E)$ to $\sim 3\%$ in the UV — the *archimedean half* (the smooth density, leading term $\frac{1}{2} \log(y/2\pi)$), an independent non-circular confirmation of the CCM density claim. But its poles sit $O(1)$ off the zeros, so the arithmetic fine-structure (the *positions*) is not captured; and the relative-determinant / Krein construction does *not* remove the Euler pole comb, because the prolate background is itself Herglotz (comb-free), leaving nothing to cancel. Both findings re-confirm Theorem 3.2: density yes, positions no; the wall is the analytic continuation. The genuine positive by-product — that the determinant channel $-D'/D$ (residues = multiplicities), not mere eigenvalue convergence, is the object that must converge ($\det_{\text{reg}}(A - z) \rightarrow C \cdot \Xi$) — sharpens the CCM frontier and is recorded among the independent results (Section 7).

3.4 Four closures of concrete constructions (not routes)

The phase produced four genuine negative results, each over a *specific constructed object or technique*. None closes a whole route; each re-types the route or names a successor.

1. **Osterwalder–Schrader reflection, natural form.** The reflection block $Q_+(x) = \int \int_0^\infty x(t) K(t+s) x(s)$ is indefinite, because the prime distributions $\delta(t+s - \log p)$

in the Weil kernel meet the half-axis $t + s = \log p$; the Toeplitz alternative $K(t - s)$ collapses to global Weil positivity = RH. The OS route in its most natural form (inversion reflection on $\{|g| \geq 1\}$) is closed-negative; another half-space choice would be needed.

2. **Möbius $\mathbb{Z}_2$ prime hub, naïve L_s^- .** An obstruction theorem: any continuation of the divergent prime traces $\text{Tr } T_s = P(s)$, $\text{Tr } T_s^2 = P(2s)$ down to $\Re s = \frac{1}{2}$ imports ζ and its zeros via $P(s) = \sum_n \mu(n)/n \cdot \log \zeta(ns)$ — hence circular. (Project-internal; not a literature-verified standard.)
3. **Toeplitz-symbol sub-method (Branch A).** A_{sin} is not Toeplitz (m -dependent oscillating diagonals $\Rightarrow$ no well-defined symbol $f(\theta)$); a structural defect of the constructed object, not merely “the method equals the open problem”. Successor: the finite Sylvester-inertia bound (done) plus GLT/outlier control.
4. **Relative-determinant / Krein bypass.** The relative determinant $-\partial_z \log[\det(A - z)/\det(A^0 - z)]$ does *not* remove the Euler pole comb: if A^0 is the prolate background it is itself comb-free (nothing to cancel, and $\det(\text{Euler} \cdot \text{prolate})/\det(\text{prolate}) = \det(\text{Euler})$ keeps the comb); the only comb-free alternative needs a self-adjoint prime factor, which is circular (= RH). This is an honest self-correction of an internally proposed “right technology”; the proposed-then-refuted arc is recorded in Section 8.

Crucially, the *determinant channel* $-D'/D$ as a *sharpening* of the CCM frontier remains valid (Section 7); only the *relative determinant* as a *bypass* of the continuation wall is refuted. The two must not be conflated.

4 Dormant Routes

A small number of constructed routes are dormant rather than dead: they are explicit reductions or partial constructions that have not been refuted, but currently lack active development. The internal register (AKTIONSPLAN.md) keeps the full inventory; the four below are summarised here because each has a direct bridge to one of the active branches.

- **D-CCM (Dirichlet–CCM operator $D_{\log}^{(\lambda, N, \chi)}$).** A direct χ -twist of the CCM rank- N perturbation construction. Addresses GRH, not RH directly — the trivial character χ_0 (Riemann case) is in this scaling regime not practically testable. Companion of Branch Z.
- **D-Twist (χ -twisted defect norm).** Diagnostic blueprint with completed Pilot v2 and Milestone 2_χ work; reformulated as λ -asymptotic statement. First tests at χ_{21}, χ_{33} ; Riemann unreachable in this scale.
- **D-PW (Paley–Wiener sector difference $\psi_\chi = \phi_\chi^+ - \phi_\chi^-$).** Classification of the five Paley–Wiener steps complete: S1, S2, S4 mechanical; S3, S5 character-specific. Continuation of the published Dirichlet Character Atlas v2 (Zenodo 20241612).
- **P-M (Möbius-graded $\mathbb{Z}_2$ prime hub).** Conditional lemma proved for finite prime set (Möbius identity exact); full prime set converges only for $\Re(s) > 1$; $\mathbb{Z}_2$ -sector statement for non-trivial zeros open. Structurally appealing alternative

interpretation of the Prime Orbit Axiom (Part I, Definition 2.1). A 2026 obstruction theorem sharpens this (Section 3.4): no non-circular L_s^- exists, since the divergent prime traces continue to $\Re s = \frac{1}{2}$ only by importing ζ and its zeros. This re-types P-M as the *prime half* of a P-M+M fusion — it provably lacks the archimedean-modular half ($s \leftrightarrow 1 - s$) that route M supplies (project-internal).

The remaining dormant entries (P-A, P-C, P-B, M, R, H) are not listed individually here; see `AKTIONSPLAN.md` for the full inventory. Two are *strategically elevated* by the criticality lens of Section 3.1: **M** (Bost–Connes $\leftrightarrow$ Meyer) and the global adelic trace route **P-A** (Connes–Consani scaling site) are of the exact-structural type and now receive additional strategic weight; **P-C** is co-resolved by an active branch; **P-B** (Berry–Keating) is of the right type but unconstructed (self-adjointness is the missing Hilbert–Pólya theorem); **R**, **H** are framework/expository.

5 External Lines

The landscape sits inside a wider research ecosystem. The following lines are cited as anchors; only the directly proof-relevant items appear here.

- **Connes 2026 Mainline** [1] (arXiv:2602.04022). Re-frames the Riemann zeros as the infrared spectrum of a self-adjoint operator and identifies the key open hypothesis MS2 (§6.6(b)). Branches A, B, Z all condition on this line.
- **Connes–van Suijlekom 2025** [3] (arXiv:2511.23257). Theorem 5.6: 1-dim even kernel of the truncated Weil form implies real zeros of $\hat{\eta}$. This is A1 = B1 in the present landscape.
- **Connes–Consani–Moscovici 2025** [2] (arXiv:2511.22755). Foundation of the rank-one perturbation $D_{\log}^{(\lambda, N)}$. Anchors Branch B’s Foundation-Lock and Branch Z’s microcluster construction.
- **Groskin 2026** (arXiv:2605.20224). Independent implementation of the CCM Galerkin matrix at sixteen cutoffs $c = 13 \dots 67, 100$ at high precision (Aitken extrapolation, $\sim 10^{-334}$ depth). Acts as external validator for branches B and Z.
- **Śliwiński 2026** (arXiv:2601.12133). Counter-regime: in the $\lambda = N$ configuration with low precision, proves an inverse-logarithmic dissonance lower bound $\epsilon(\lambda, N) \geq 1/(4 \ln \lambda)$. Does not bound the fixed- λ , $N \gg \lambda$, high-precision setup of this series; stands as a warning against naïve $\lambda = N$ limits.
- **Watson 2026** (arXiv:2602.01248), “The Riemann Ξ -function from primitive Markovian cycles I.” Constructs a Laguerre–Pólya function Ψ from finite reversible Markov dynamics, “entirely Archimedean, using no Euler products, primes, or arithmetic spectral input”; not an RH proof — it isolates “a single remaining analytic problem relating Ψ to $\Xi(2 \cdot)$.” An independent triangulation of the common wall: from the archimedean side it reaches the same *identification* step (relating the constructed object to Ξ) that the routes of this series reach from the arithmetic side (Section 3.2).

- **Franchi Viceré 2026** (Zenodo 19546495). Claims unconditional RH proof via apeirohedron / semi-local spectral descent. Surveyed and found not viable: the stabilisation argument confuses pointwise with strip-uniform convergence; the Apeirohedron limit step is not rigorous; boundary conditions do not replace the even-sector / MS1 input. See `AKTIONSPLAN.md §Klasse 8`.

6 Systematically Explored Alternatives

Eleven alternative strategies (BI-1 through BI-11) were systematically explored during development. This exhaustive search is itself a meta-result: the solution space is mapped.

Approach	Core idea	Result	Status
BI-1: $\det_2$ / Carleman	Extend to Hilbert–Schmidt class	No positivity substitute	Refuted
BI-2: RG fixed point	Weil positivity under scaling	Not formalizable	Discarded
BI-3: Universal property	Physical plausibility $\Rightarrow$ RH	Too abstract	Discarded
BI-5: Lifting programs	Weil $\rightarrow$ Grothendieck $\rightarrow$ RMT	All open except $\mathbb{F}_q$	Not usable
BI-6: Implication ring	$Li \leftrightarrow Weil \leftrightarrow NB \leftrightarrow BD$	Ring does not exist	Refuted
BI-7: Pöschl–Teller scattering	Gamma structure analog to ξ	No off-line stability	Negative
BI-8: Second-order stability	RH as variational problem	Architecture correct	Realized
BI-9: Soliton models	Topological stability	Coulomb gas unstable	Refuted
BI-10: Building blocks	No-Coordination, shift structure	Successfully reused	Integrated
BI-11: Connes 2026	Even dominance via Q_W	Breakthrough	Core path

The chosen path (BI-8 + BI-10 + BI-11) combines the variational architecture, the arithmetic building blocks, and Connes’ spectral framework. Routes A and B (total positivity, commuting operator) were ruling-outs that are structural discoveries: even dominance holds *despite* their absence, which clarifies that the mechanism is genuinely arithmetic-statistical rather than algebraic (Part II [5]).

7 Independent Results

Several results from this program are of independent interest, regardless of the main proof’s status:

Result	Context	Significance	Status
<i>Positive results</i>			
Shift Parity Lemma	Arithmetic trigonometry	Every prime favors even eigenfunctions	Proved
No-Coordination Lemma	Multiplicative independence	Arithmetic substitute for entropy term	Proved
Leading-Mode Cancellation ($c = 2 + \sqrt{2}$)	Resolvent analysis	Universal pairwise cancellation constant	Proved
Euler–Maclaurin: $\rho^{\text{EM}} < 0$	PNT asymptotics	Global stability anchor	Proved
33 finite-range CAP diagnostics	Interval arithmetic	Conditional bridge	Evidence
Concentration lemma	Even-dominance image measure	No fixed prime set controls a positive share of prime mass	Proved
Finite Sylvester inertia ($\nu_- = 0$)	Interval arithmetic, $\lambda = 100$	Finite even dominance rigorous, no symbol needed	Verified
Residue-calibration channel	Weyl-function diagnostic	Eigenvalue conv. $\neq$ Weyl conv.; channel is $-D'/D$, $\det_{\text{reg}} \rightarrow C\Xi$	Diagnostic
<i>Negative results</i>			
Obstructions K1–K4	Dead-end analysis	Maps impossible strategies	Documented
No absolute PF_∞ for A_λ	Prime shift multiplier	$M(\xi)$ negative on $\sim 49\%$ of domain	Proved
No universal commuting operator	Sturm–Liouville search	Kernel of $[T, D_N(r)]$ is $\{c \cdot I\}$	Proved
<i>Structural insights</i>			
Inverted-Hessian structure	YM/TU/RH comparison	RH is a maximum, not minimum	Confirmed
Cross-disciplinary catalog	Regularization	11 fields share trace-class obstruction	Documented

The negative results protect future researchers from repeating dead-end strategies. The Shift Parity Lemma, the obstruction analysis (K1–K4), and the Mellin decay barrier are publishable independently.

8 Hypothesis Corrections in the Iterative Process

The iterative development produced several hypothesis corrections that are documented here as a record of honest research practice (Sammler-Disziplin).

8.1 GF1 Correction Chain (Iterations 15–29)

Iter	Hypothesis	Status
15	$C_m \approx 8$ universal	falsified (boundary effect)
18	$C_m(\lambda = 5) \approx 4.3$	falsified (no plateau)
19	$\text{sLI} \sim 1/N$	confirmed, but without λ -scaling
20	$N \cdot \text{sLI} \approx 329$ universal	falsified
27	$\text{sLI} = \mathbf{C}^* \lambda / (\mathbf{NL}) + \mathbf{o}(\cdot)$	confirmed for $\lambda = 5, 7$, $C^* = 64\pi^2/3$
28	$\lambda = 3$ all- λ check	low-bandwidth outlier; two-regime guardrail
29	$\lambda = 11$ discriminator and endpoint-kernel audit	$C_{\text{eff}} \approx 168$ at $N = 200$; simple endpoint factorisation falsified

8.2 B-1 Corrections and c_{Ξ} RH-Neutral Form

The Hadamard constant c_{Ξ} was initially written in the RH-specialised form $\sum_{k \geq 1} 1/\gamma_k^2$ (sum over imaginary parts of zeros on the critical line). An external audit corrected this to the RH-neutral form $c_{\Xi} = -\frac{1}{2} \sum_{\rho} 1/(\rho - \frac{1}{2})^2$ (sum over all non-trivial zeros). This is a Sammler-Disziplin correction: the RH-neutral form is mathematically more general and does not assume what is being proved.

8.3 The Relative-Determinant Bypass: Proposed and Refuted (2026)

During the 2026 phase, the relative determinant (Krein spectral shift) $-\partial_z \log[\det(A - z)/\det(A^0 - z)]$ was proposed internally as the “right technology” to remove the upper-half-plane Euler pole comb and so bypass the continuation wall (Theorem 3.2). A subsequent test on the genuine Connes–Moscovici prolate background refuted this as a bypass: the prolate A^0 is itself Herglotz (comb-free), so there is nothing to cancel ($\det(\text{Euler} \cdot \text{prolate})/\det(\text{prolate}) = \det(\text{Euler})$ retains the comb), and the only comb-free alternative needs a self-adjoint prime factor, which is circular (= RH). The proposal is withdrawn as a bypass. The *separate* finding — that the determinant channel $-D'/D$, not mere eigenvalue convergence, is the correct object to converge ($\det_{\text{reg}}(A - z) \rightarrow C \Xi$) — remains valid (Section 7); the two must not be conflated.

A second correction: an internal phase lemma $\Theta_{\sigma} = \xi(\sigma - iz)/\xi(\sigma + iz)$ was sign-reversed (its poles come from the denominator, so under RH it *has* poles); the correct object is the reciprocal phase $\hat{\Theta}_{\sigma} = \xi(\sigma + iz)/\xi(\sigma - iz)$, with $\text{RH} \Leftrightarrow \hat{\Theta}_{\sigma}$ pole-free in the upper half-plane for all $\sigma > \frac{1}{2}$.

8.4 Value of the Correction Record

These corrections demonstrate that the research process is self-correcting. Each correction sharpened the theoretical framework: the GF1 chain moved from a naïve constant ($C_m \approx 8$) through several intermediate forms to the $\lambda = 5, 7$ boundary-moment benchmark $\text{sLI} \sim C^* \lambda / (NL)$; the $\lambda = 3$ sweep and the open $\lambda = 11$ discriminator prevent an all- λ theorem reading, and the c_{Ξ} correction removed a circular RH-dependence from a key constant.

9 Update (v3.2): The RH/abc Archimedean-Mass Crosswalk

A cross-problem line that had been carried informally alongside the routes of this series was audited to a close between June and August 2026. It is recorded here in Part IV rather than with the spectral material of Part III [7], because it concerns a bridge between RH and a second conjecture rather than a route inside the RH landscape itself.

The crosswalk (register designation B2, state V1.1) compared the archimedean mass objects on the two sides: the RH-side μ_- , a signed dimensionless negative mass, against the abc-side Ω_E , a positive Néron period. Only a structural analogy is confirmed — the two are not instances of a common object, and no shared normalisation emerged under which they could be compared quantitatively. An accompanying candidate audit examined four proposals for such a common object, namely the Connes–Consani trace, the Yuan–Zhang height pairing, the Frey-period log-height, and a formal container, and found that none of them satisfies the four bridge obligations. The audit rests on five bound sources with producer and verifier quotas of 19/19 each, fail-closed. The consequence for the landscape is that the $\text{RH} \leftrightarrow \text{abc}$ bridge is exhausted *as a linguistic analogy*: a successor would have to supply a normalised common H_∞ with typed support, both realisations, compatible normalisations, and two non-circular inequalities. Claim boundary: what closes here is the candidate list examined and the analogy reading of the bridge — neither conjecture is refuted, and an object outside the four tested candidates is untested rather than excluded.

10 Summary

Part IV contributes the cross-route perspective: the three active branches share a Galerkin pattern (fixed-parameter positive, uniform-limit open), and two isolated proof slots (B-1 log-curvature, GF1 factor-16 identification) are the concrete analytical targets, which remain well-posed open problems. Section 3 then adds a 2026 *heuristic* criticality lens ($\text{RH} \Leftrightarrow \Lambda = 0$) that re-weights the routes toward exact-structural mechanisms (without refuting the margin-type slots), localizes the common wall as the analytic continuation across $\Re s = 1$, records a computable Laguerre–Pólya/Herglotz diagnostic (a classical criterion turned into a test instrument), and reports four closures of *concrete constructions* — the OS reflection in natural form, the naïve Möbius L_s^- , the Toeplitz-symbol sub-method, and the relative-determinant bypass — none of which closes a route or RH. The dormant and external routes are catalogued for completeness; the hypothesis correction record provides transparency. The condensed atlas and final assessment follow in Part V [4].

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tributed to conceptual exploration, analytical work, literature review, writing, and critical review. The author, Lukas Geiger, conceived the research direction, provided domain expertise, and exercised final editorial authority. The author assumes full and sole scholarly responsibility for all content, including any errors.

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